

Logos Experiment 18 Report

Large Scaling + 15k Training with Weight Tying + SwiGLU

Experiment 18 extends the best large Logos architecture from previous runs to 15,000 training iterations. The goal is to test whether the 124M parameter decoder-only model continues improving beyond the 10,000-iteration result from Experiment 17.

1. Experiment Setup

Field	Value
Run name	logos_exp18_large_scaling_weight_tying_swiglu_15k_train
Architecture	Decoder-only GPT-style language model
Dataset	OpenWebText subset, 100k samples
Tokenizer	GPT-2 tokenizer
Parameters	124,031,232
Layers	12
Embedding dimension	768
Attention heads	12
Head dimension	64
Context length	512
Batch size	32
Tokens per step	16,384
Feed-forward network	SwiGLU
Weight tying	Enabled
Training iterations	15,000

2. Final Results

Metric	Result
Best iteration	15,000
Best validation loss	3.9551
Final validation loss	3.9811
Final validation perplexity	53.57
Checkpoint size	473.20 MB
Peak VRAM	17.73 GB
Training time	56.80 minutes
Generation speed	109.73 tokens/sec
Generation latency	9.11 ms/token
Average GPU power	293.19 W
Estimated GPU energy	277.39 Wh

3. Comparison with Experiment 17

Metric	Exp 17: 10k iterations	Exp 18: 15k iterations
Parameters	124.03M	124.03M
Dataset	100k OpenWebText	100k OpenWebText
Context	512	512
Best validation loss	4.1182	3.9551
Final validation loss	4.1167	3.9811
Final perplexity	61.36	53.57

Checkpoint	473.20 MB	473.20 MB
Peak VRAM	17.73 GB	17.73 GB
Training time	37.73 min	56.80 min
Energy	183.67 Wh	277.39 Wh

4. Scientific Interpretation

Experiment 18 is the strongest Logos model so far. Increasing training from 10,000 to 15,000 iterations improved final validation loss from 4.1167 to 3.9811 and reduced perplexity from 61.36 to 53.57 without changing model size, checkpoint size, context length, or peak VRAM. This supports the conclusion that the 124M parameter model was still learning after Experiment 17.

The cost is higher training time and energy. Training time increased from 37.73 minutes to 56.80 minutes, while estimated GPU energy increased from 183.67 Wh to 277.39 Wh. This makes Experiment 18 the best quality model, but not the most energy-efficient checkpoint.

The best training-log validation loss was 3.9551, while the final re-evaluated validation loss was 3.9811. This difference is expected because the validation estimator samples random batches. For summary tables, final validation loss and perplexity should be treated as the main benchmark values, while best validation loss should be reported as the best sampled checkpoint estimate.

5. Current Status

Current best model: Experiment 18. It is a 124.03M parameter decoder-only GPT-style language model trained on 100k OpenWebText samples using GPT-2 tokenization, 512-token context length, SwiGLU feed-forward layers, and weight tying. It achieved final validation loss 3.9811 and final perplexity 53.57.

Next recommended actions: update the paper-level results table, consider whether to run a plateau test at 20k iterations, and begin adding a formal resource-aware efficiency metric or quantization experiment for a stronger novelty contribution.